

OR2 — Architecture Document

NLU and Diagnostic Engine | AI vs Cancer — DermaTriage

OR2 — Architecture Document

NLU and Diagnostic Engine

1. System Overview

Project	AI vs Cancer — DermaTriage
Purpose	AI-assisted dermatology triage for early skin cancer detection
Institution	University of Milan
Supervisor	Prof. Ernesto Damiani
Developer	Basil
Platform	B4 (Social Things)
Team	Ricardo Matamoros, Niccolo
Live URL	https://5000--main--a14-cancer--basybasy00.coder.ricerca.sesar.di.unimi.it
B4 Portal	https://aivscancer.b4service.it
GitHub	github.com/basil-github/image_classifier

2. Four-Stage AI Pipeline

DermaTriage processes patient-submitted skin lesion images through a sequential four-stage AI pipeline, each stage enriching the clinical context before the final triage decision.

Stage	Model	Input	Output
1	EfficientNet-B4 (fine-tuned)	Skin lesion image	HIGH / MEDIUM / LOW + confidence
2	Qwen2-VL-7B-Instruct	Skin lesion image	Clinical text description (5-bullet format)
3	RAG — ChromaDB	Clinical text	Top-5 similar historical cases
4	BioMistral-7B	Stage 1+2+3 + symptoms	JSON: urgency, confidence, reasoning, pathology

Pipeline Flow:

1. Patient uploads image + age + sex + localization via B4 chatbot
2. Stage 1: `predict_image()` — EfficientNet classifies urgency (threshold: `prob >= 0.25` for HIGH)
3. Stage 2: `_call_qwen2()` — Vision LLM generates 5-bullet clinical description
4. Stage 3: `find_similar_cases()` — ChromaDB retrieves top-5 similar cases by cosine similarity
5. Stage 4: `_call_biomistral()` — Medical LLM synthesises final JSON decision
6. Adaptation Layer: `map_urgency_to_p_scale()` assigns P1-P4, specialist and SLA
7. Result saved to B4 via `b4_client.save_prediction()` and locally via `save_diagnosis()`

Adaptation Layer Mapping:

Urgency + Condition	P-Scale	SLA
HIGH + confidence > 0.85	P1	24 hours
HIGH	P2	48 hours
MEDIUM	P3	72 hours

OR2 — Architecture Document

NLU and Diagnostic Engine | AI vs Cancer — DermaTriage

Urgency + Condition	P-Scale	SLA
LOW	P4	7 giorni

Fallback: When no image is available, symptom-only scoring derives urgency from B4 chatbot interaction fields (itching, bleeding, growing, changing, pain, etc.).

3. API Endpoints

3.1 DermaTriage API (20 endpoints)

Method	Path	Auth	Description
GET	/	None	Home page
GET	/triage	None	Triage upload page
POST	/analyze	None	Full 4-step pipeline (direct upload)
POST	/diagnose	X-API-Key	B4-integrated pipeline
GET	/health	None	Service health check
GET	/stats	None	Pipeline statistics
GET	/admin/diagnoses	X-API-Key	Diagnoses dashboard
GET	/admin/review/{id}	X-API-Key	Doctor validation page
POST	/admin/validate/{id}	X-API-Key	Save doctor correction
GET	/admin/prompts	X-API-Key	Prompt manager dashboard
POST	/admin/prompts/update	X-API-Key	Rebuild prompts from corrections
POST	/admin/prompts/reset/{model}	X-API-Key	Reset prompt to default
GET	/admin/prompts/history/{model}	X-API-Key	Prompt version history
GET	/admin/retraining	X-API-Key	Retraining dashboard
GET	/admin/retraining/status	X-API-Key	Model version + correction progress
POST	/admin/retraining/check-and-trigger	X-API-Key	Event-driven retraining trigger
POST	/admin/retraining/run	X-API-Key	Manual retraining trigger
GET	/admin/retraining/history	X-API-Key	Version history list
POST	/admin/retraining/rollback	X-API-Key	Restore previous model
GET	/admin/retraining/fetch	X-API-Key	Pull validated outcomes from B4

3.2 B4 External API Calls

Method	B4 Path	Purpose
POST	/api/v1/auth/token	JWT authentication
GET	/api/v1/triage/list	Fetch all consultos
GET	/api/v1/chatbot/consulto/{id}/documenti	List documents
GET	/api/v1/chatbot/documento/{id}	Download image
PUT	/api/v1/chatbot/consulto/{id}/diagnostic-output	Save AI result
PUT	/api/v1/chatbot/consulto/{id}/medical-validation	Save doctor validation

OR2 — Architecture Document

NLU and Diagnostic Engine | AI vs Cancer — DermaTriage

Method	B4 Path	Purpose
GET	/api/v1/chatbot/consulto/{id}/validated-outcome	Read validated outcome
GET	/api/v1/chatbot/health	B4 health check

4. Self-Learning Loops

4.1 Prompt Evolution

Parameter	Value
Manager	PromptManager (singleton, thread-safe)
Storage	db/prompts.json
Trigger	Every 10 doctor corrections
Method	Sliding window of 20 most recent examples
BioMistral version	v4 (3 updates, last: 2026-04-30)
Qwen2-VL version	v3 (2 updates, last: 2026-04-30)

4.2 CNN Retraining Loop

Parameter	Value
Trigger	50 doctor corrections where agrees == False
Source	GET /api/v1/triage/list
Image download	GET /api/v1/chatbot/documento/{upload.id}
Label mapping	P1/P2 -> HIGH, P3 -> MEDIUM, P4 -> LOW
Fine-tune	Last 2 EfficientNet feature blocks + classifier head
LR backbone	5e-6
LR classifier	3e-5
Epochs per cycle	10
Deploy condition	New sensitivity >= old AND false_low <= old AND accuracy >= old - 5%
Rollback	Automatic if accuracy drops > 5%
Trigger endpoint	POST /admin/retraining/check-and-trigger (event-driven, NOT cron)

5. Storage Architecture

Component	Location	Size
EfficientNet weights	models/efficientnet_b4.pth	72 MB
ChromaDB vector store	chroma_db/	~15.3 MB
RAG dataset	data/RAG_dataset.csv	~5 MB
Diagnoses history	db/diagnoses.json	Flat JSON
Doctor corrections	db/corrections.json	Flat JSON
Prompt versions	db/prompts.json	Flat JSON

OR2 — Architecture Document

NLU and Diagnostic Engine | AI vs Cancer — DermaTriage

Component	Location	Size
Model versions	db/model_versions.json	Flat JSON
Training images	data/images/	~2 GB
Logs	logs/	Rotating

6. Technology Stack

Category	Technology	Version
Web Framework	FastAPI + Uvicorn	Latest
CNN Model	EfficientNet-B4 (PyTorch)	Custom fine-tuned
Vision LLM	Qwen2-VL-7B-Instruct	HuggingFace
Medical LLM	BioMistral-7B	HuggingFace
Embeddings	all-MiniLM-L6-v2	sentence-transformers
Vector Store	ChromaDB	Latest
HTTP Client	httpx	Async
Image Processing	Pillow	Latest
Data Processing	pandas, numpy	Latest
Balancing	imbalanced-learn (SMOTE)	Latest
Python	CPython	3.12.3
GPU	NVIDIA L40S	CUDA enabled
PyTorch	PyTorch	2.11.0

7. Live System Status

Service	Status	Notes
EfficientNet-B4	Running	Local model, 72MB
ChromaDB	Running	2,298 records indexed
B4 Connection	Running	JWT auto-refresh active
Ollama	Not used	BioMistral via HuggingFace directly

Metric	Value
Total diagnoses processed	30
Doctor corrections	1
Urgency distribution	P1: 2, P2: 0, P3: 24, P4: 4